

Fair Cost Allocation in Energy Communities: A DLMP-based Bilevel Optimization with a Scalable Shapley Value Approach

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Abstract—Energy communities (ECs) are emerging as a promising decentralized model for managing cooperative distributed energy resources (DERs). As multiple ECs become increasingly integrated into distribution system operation, ensuring fairness in allocating operating costs among communities becomes a challenge. Distribution network constraints can affect ECs differently depending on their locations, for example, by causing unequal levels of solar generation curtailment. These disparities can lead to unfair inter-community cost distribution, particularly when Distribution Locational Marginal Prices (DLMPs) are employed. This highlights the need for cost-allocation mechanisms that account for both network constraints and community-level impacts. Despite growing interest in fairness-aware methods, most methods do not account for the impact of DLMPs. To fill this gap, we propose a bilevel optimization model in which a Community Energy Aggregator (CEA) schedules DERs across multiple ECs while a Distribution System Operator (DSO) determines DLMPs through network-constrained dispatch. Leveraging the Karush-Kuhn-Tucker (KKT) conditions and strong duality, the bilevel model is reformulated into a tractable single-level problem. We achieve fairness in the cost allocation by applying the Shapley value to quantify each community's marginal contribution to system-wide cost savings. The effectiveness of the proposed method is validated through simulations on several benchmark distribution systems.

Index Terms—Fairness-aware, Energy community, LinDistFlow, DLMP, Shapley value

NOMENCLATURE

Sets

$\mathcal{N}$	Set of all distribution buses indexed by n .
$\mathcal{N}_c$	Set of buses where ECs are located, $\mathcal{N}_c \subseteq \mathcal{N}$.
$\mathcal{N}_s$	Set of slack buses (single bus at index 0).
$\mathcal{M}$	Set of ECs considered in the allocation problem, indexed by m .
$\mathcal{C}$	Coalition of ECs, where $\mathcal{C} \subseteq \mathcal{M}$.
$\mathcal{L}$	Set of distribution lines indexed by (i, j) .
$\mathcal{T}$	Set of time periods indexed by t .

Parameters

$\hat{P}_{n,t}^{\text{pv}}$	Forecasted available PV generation.
$\text{PF}_{\min}$	Minimum allowable power factor of PV inverter.
ζ^{pv}	Reactive power coefficient derived from $\text{PF}_{\min}$.
$\bar{P}_n^{\text{ch}}, \bar{P}_n^{\text{dis}}$	Charging/discharging power capacities of BESS.

$\underline{E}_n, \bar{E}_n$	Min/max energy limits of BESS.
$\eta_n^{\text{ch}}, \eta_n^{\text{dis}}$	Charging/discharging efficiencies of BESS.
$\bar{P}_{n,t}^{\text{load}}, \bar{Q}_{n,t}^{\text{load}}$	Forecasted active and reactive loads.
π	Compensation rate for curtailed load.
r_{ij}, x_{ij}	Resistance and reactance of line (i, j) .
z_{ij}	Impedance of line (i, j) , i.e., $z_{ij} = r_{ij} + jx_{ij}$.
π_t^{LMP}	Wholesale electricity price.
c_n^g	Marginal cost of local generator.
$\underline{p}_n^g, \bar{p}_n^g$	Active power output limits of generator.
$\underline{q}_n^g, \bar{q}_n^g$	Reactive power output limits of generator.
$v^{\min}, v^{\max}$	Lower/upper limits of bus voltage magnitude.
Δt	Duration of each time period.

Decision Variables

$p_{n,t}^{\text{pv}}, q_{n,t}^{\text{pv}}$	Active/reactive power outputs of PV system.
$p_{n,t}^{\text{ch}}, p_{n,t}^{\text{dis}}$	Charging/discharging power of BESS.
$b_{n,t}^{\text{ch}}, b_{n,t}^{\text{dis}}$	Binary indicators for charging/discharging mode of BESS.
$e_{n,t}$	State of charge of BESS.
$p_{n,t}^{\text{load}}$	Scheduled controllable load.
$p_{n,t}^c, q_{n,t}^c$	Active/reactive power consumption of the community.
$p_{n,t}^g, q_{n,t}^g$	Active/reactive power outputs of the generator.
$u_{n,t}$	Squared voltage magnitude.
$P_{ij,t}, Q_{ij,t}$	Active/reactive power flows on line (i, j) .
$p_{n,t}^{\text{inj}}, q_{n,t}^{\text{inj}}$	Net active/reactive power injection.

Lagrange Multipliers

$\lambda_{n,t}^p, \lambda_{n,t}^q$	Multipliers for active and reactive power balance; interpreted as active and reactive DLMPs.
$\lambda_{ij,t}^{\text{volt}}$	Multiplier for voltage drop constraint on line (i, j) .
$\lambda_{n,t}^u, \bar{\lambda}_{n,t}^u$	Multipliers for lower and upper bounds on squared voltage magnitude.
$\lambda_{n,t}^{gp}, \bar{\lambda}_{n,t}^{gp}$	Multipliers for lower and upper bounds on $p_{n,t}^g$.
$\lambda_{n,t}^{gq}, \bar{\lambda}_{n,t}^{gq}$	Multipliers for lower and upper bounds on $q_{n,t}^g$.

I. INTRODUCTION

A. Background and Motivation

The rapid proliferation of prosumers equipped with renewable energy resources (RES) and battery energy storage systems (BESS) has significantly increased interest in decentralized energy systems in recent years [1]. One such example is the rise of energy communities (ECs) [2], which coordinate distributed energy resources (DERs) and flexible loads at the local level. A real-world example of such setup is the EcoGrid 2.0 project, where grid-connected resources were coordinated to provide flexibility services to the distribution system operator (DSO) [3]. While ECs offer various technical and economic benefits, such as improved self-consumption, peak shaving, and reduced grid dependency, their effective

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formation and operation pose several challenges. A critical issue lies in the fair allocation of collective operating costs and benefits arising from shared assets or coordinated operation within the distribution grid [4]. This concern is also reflected in European regulations, which require renewable ECs to face cost-reflective network charges and to contribute to overall system cost sharing in an adequate, fair, and balanced manner as mentioned in the EU directive 2023/2413 [2].

In this context, recent studies have increasingly explored fair allocation mechanisms. For example, [5] adopts a Nash bargaining-based priority mechanism for peer-to-peer energy trading in local communities, and a nucleolus analysis is used in [6] to ensure coalition stability in transmission expansion planning models. The Shapley value [7] has also been widely used, providing a contribution-based approach to fair cost and benefit allocation. Under this approach, the total cost (or benefit) is distributed according to each participant's marginal contribution across all possible coalitions and satisfies key fairness properties, including efficiency, symmetry, and the null player property [8]. These characteristics make it a transparent solution for cooperative cost sharing, particularly when multiple parties share infrastructure or coordinate operations.

Building on this idea, several studies have applied the Shapley value in the context of investment allocation for ECs. For instance, the work in [9] proposed a Shapley-based incentive scheme to allocate the cost of grid-enhancing technologies fairly among multiple participants. Similarly, [10] introduced a scalable Shapley value method to determine fair investment strategies for large energy communities, focusing on equitable distribution of infrastructure costs such as shared batteries or RES installations. The Shapley value has also been incorporated into the design of market mechanisms [11], [12]. Other work has expanded these fairness schemes to be equitable by considering socio-economic aspects such as income levels and capability to purchase DERs [13].

Moreover, in future distribution systems, bus-level electricity prices are proposed through Distribution Locational Marginal Prices (DLMPs) [14], which reflect the spatiotemporal value of energy and grid constraints. DLMP concepts have been widely studied as pricing signals but have not been applied to fair cost allocation among ECs. In practice, each EC's operation can influence DLMPs through its net injections and location. Thus, an EC's scheduling decisions affect both its own cost and system-wide DLMPs and cost distribution.

To show the impact of DLMP-based pricing on fairness, let us consider a 3-bus toy example in Fig. 1. Bus 0 is the slack bus connected to the upstream grid, while Communities A and B are connected at buses 1 and 2, respectively. The active power flows from bus 0 to bus 1 and from bus 1 to bus 2 are denoted by P_{01} and P_{12} , with corresponding line resistances

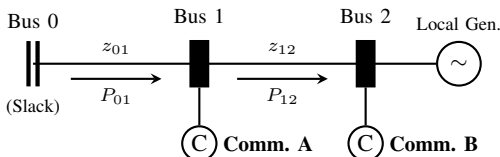


Fig. 1. The 3-bus radial distribution feeder used in the motivating example. A voltage lower bound is imposed only at the downstream bus (bus 2).

TABLE I
COST-BENEFIT ALLOCATION UNDER DLMP-BASED SETTLEMENT

	Community A (Bus 1)	Community B (Bus 2)
Action	None	Injection Δ
Injection cost	0	$c_{\Delta}\Delta$
DLMP change	$r_{01}\Delta\mu$	$(r_{01} + r_{12})\Delta\mu$
Share of total benefit [†]	$d_1 r_{01}$	$d_2(r_{01} + r_{12})$
System contribution	$d_1 r_{01} + d_2(r_{01} + r_{12})$	$d_1 r_{01} + d_2(r_{01} + r_{12})$
	0	100%

[†] Benefit is defined as the reduction in DLMP-based payment, i.e., $d_i |\Delta\lambda_i|$.

r_{01} and r_{12} . Let u_i denote the squared voltage magnitude at bus i , where u_0 is fixed at the slack bus and $\underline{u}$ is the lower voltage limit. The fixed demands at buses 1 and 2 are denoted by d_1 and d_2 , respectively. In this example, we assume that Community B can provide a controllable net injection $\Delta \geq 0$, e.g., through battery discharge or flexible demand reduction, so that the net demand at bus 2 becomes $d_2 - \Delta$. The DSO minimizes the total generation cost subject to nodal power balance and a voltage constraint:

$$\begin{aligned}
 \min_{p_0^g, p_2^g, P_{01}, P_{12}} \quad & c_0 p_0^g + c_2 p_2^g \\
 \text{s.t.} \quad & p_0^g - P_{01} = 0 & : \lambda_0 \\
 & P_{01} - d_1 - P_{12} = 0 & : \lambda_1 \\
 & P_{12} + p_2^g - (d_2 - \Delta) = 0 & : \lambda_2 \\
 & u_2 \geq \underline{u} & : \mu \geq 0 \\
 & 0 \leq p_2^g \leq \bar{p}_2^g,
 \end{aligned} \tag{1}$$

where p_0^g denotes the power supplied from the upstream grid at the slack bus, and p_2^g denotes the output of the local generator at bus 2. The corresponding marginal costs are c_0 and c_2 , respectively, with $c_2 > c_0$ indicating that the local generator is more expensive than the upstream supply.

Using the LinDistFlow approximation [15], the squared voltage magnitude at bus 2 is expressed as $u_2 = u_0 - 2[r_{01}d_1 + (r_{01} + r_{12})(d_2 - \Delta)]$. The voltage constraint $u_2 \geq \underline{u}$ can therefore be rewritten as $r_{01}d_1 + (r_{01} + r_{12})(d_2 - \Delta) \leq \frac{1}{2}(u_0 - \underline{u})$. It reflects that the demand at bus 1 contributes to the voltage drop through the line between buses 0 and 1, whereas the net demand at bus 2 contributes to voltage drop through both lines. As $r_{01} + r_{12} > r_{01}$, bus 2 injections are more effective in relieving the voltage constraint compared to bus 1 injections.

The DLMP at each bus is the Lagrange multiplier associated with the corresponding bus power balance constraint. From the KKT conditions of (1), the slack-bus price satisfies $\lambda_0 = c_0$, and the DLMPs are $\lambda_1 = \lambda_0 + r_{01}\mu$, $\lambda_2 = \lambda_0 + (r_{01} + r_{12})\mu$. When the voltage constraint is inactive ($\mu = 0$), the prices are uniform, i.e., $\lambda_1 = \lambda_2 = c_0$. However, when the voltage constraint becomes binding ($\mu > 0$), the DSO must dispatch the more expensive local generator at bus 2, which raises the DLMPs at both buses, with a larger increase at bus 2.

Now consider an increase in the injection Δ by Community B. This action relaxes the downstream voltage constraint, reducing the associated shadow price μ . As a result, the DLMPs at both buses decrease: $\Delta \uparrow \Rightarrow \mu \downarrow \Rightarrow \lambda_1 \downarrow, \lambda_2 \downarrow$.

This example reveals a cross-community externality under DLMP-based settlement. As shown in Table I, Community B bears the full cost of the corrective action, while the resulting DLMP reduction benefits both communities, including

Community A, which does not change its operation. This highlights a fairness challenge for a CEA managing multiple communities and motivates the use of a contribution-based mechanism, such as the Shapley value.

Another challenge with the Shapley-based method is their high computational complexity for their application in large-scale systems [4], [16]–[21]. To address this, approximation methods based on statistical sampling have been proposed, such as the stratified sampling highlighted in [16]. Others suggest grouping or clustering players to reduce the problem’s dimensionality. However, the criteria for grouping in these studies are typically based on players’ operational and economic characteristics, such as consumption patterns [4], their energy profiles [17], or market impact [18]. While these approaches reduce complexity, they do not directly account for the underlying physical characteristics of the power grid, which fundamentally determine the players’ interactions.

B. Contributions

In light of these gaps, this paper proposes a fair cost allocation framework for a Community Energy Aggregator (CEA) coordinating multiple DER-equipped ECs under DLMP-based pricing. The problem is formulated as a bilevel optimization problem accounting for the interaction between the CEA and the DSO. The scheme determines the DLMPs, which are then used to determine fair cost allocation using the Shapley value. To improve the scalability of Shapley value computation in large-scale systems, we propose a signature-based approximation scheme. The proposed framework can offer valuable policy insights into (i) fairness-aware community coordination, (ii) incentive-compatible market design, and (iii) mechanisms to encourage DER participation in distribution-level markets.

The key contributions of this work are listed below.

- (i) We develop a **tractable single-level reformulation of the conventional bilevel DLMP-based scheme** using the Karush-Kuhn-Tucker (KKT) conditions and strong duality, enabling numerical solution under realistic settings.
- (ii) We propose a **Shapley value-based fair cost allocation** method that redistributes each community’s marginal contribution using DLMPs, and we evaluate its performance through case studies on benchmark distribution networks.
- (iii) We propose a **signature-based approximation** method for **scalable Shapley value computation** based on their physical and topological characteristics.

Compared with existing fairness-aware studies on energy communities (ECs) [4], [21]–[23], which consider static pricing in the cost-benefit sharing, this work explicitly incorporates DLMPs into the cooperative framework. In contrast to fair photovoltaic (PV) curtailment approaches [24], [25], which consider network constraints through curtailment policies or network reconfiguration strategies, our study focuses on the allocation of cooperative cost savings arising from coordinated EC operation. Furthermore, unlike the bilevel framework proposed in [26], which is mainly concerned with hosting-capacity analysis, this work addresses the fair distribution of cooperative cost savings in settings where EC coordination influences DLMPs and overall system operating costs. To this end, we integrate DLMP-based operational optimization

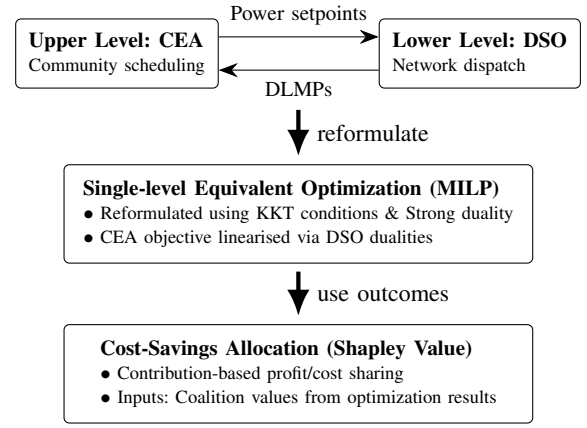


Fig. 2. Proposed framework: Bilevel interaction, single-level reformulation, and Shapley-based cost-savings allocation.

with a Shapley-value-based allocation mechanism, providing a fairness-aware approach that accounts for both network conditions and economic benefits.

The remainder of this paper is organized as follows. Section II describes the operational policies of the CEA and the DSO, including the problem structure and mathematical formulation. Section III presents the coalitional game-theoretic model and details the Shapley value method for fair cost allocation among communities. In Section IV, case studies are provided to evaluate the proposed methods, and Section V summarizes the findings and concludes the paper.

II. CEA & DSO SCHEDULING FRAMEWORK

A. Description of Problem Structure

Our goal is to coordinate the operation of different local communities and the distribution network. This is modeled using a bilevel optimization framework, where the CEA and the DSO are structured as the leader and the follower, respectively. The overall structure of this scheme is illustrated in Fig. 2. The upper-level problem represents the CEA’s decision-making on community-level energy scheduling, while the lower-level problem reflects the DSO’s response in maintaining system-wide balance and operational feasibility.

The CEA optimizes the operation of the ECs connected to the distribution grid, while the DSO is responsible for the reliable operation of the distribution network. We assume that each EC hosts various DERs (e.g., PV, BESS, and controllable loads) that are directly controlled and optimized by the CEA. The DSO procures electricity at the transmission-distribution interface at wholesale market prices, supplies power to both non-community consumers and ECs, and controls local generators while respecting distribution network constraints. Although CEA scheduling is similar to aggregator operation of distributed microgrids in terms of DER coordination, the ECs in this paper are connected to the same distribution network, with the main focus on allocating DLMP-induced cooperative cost savings among ECs.

The price signals used in CEA scheduling are DLMPs derived from the DSO’s optimization problem. These DLMPs are the dual variables associated with active and reactive power balance constraints at each bus. Consequently, the cost associated with each EC is affected by the locational and temporal conditions of the distribution grid. At the same time, the

CEA's scheduling decisions influence the bus power injections affecting the DSO's power balance and the resulting DLMPs. This interdependency results in a coupled bilevel optimization problem. The CEA and DSO models are described next.

B. CEA Model: Upper Level Problem

The objective of the CEA is to minimize the total cost of active and reactive power consumption under DLMP-based price signals (from DSO), along with the incentive payments for flexible load reductions. The CEA schedules the DERs and flexible loads of the participating ECs subject to their operational constraints. For notational simplicity, the formulation represents EC operation at the bus level. If multiple ECs are connected to the same bus, their net consumption can be aggregated because as they face the same DLMPs. The resulting bus-level cost or savings can then be further allocated among the collocated ECs through a separate post-settlement mechanism. The CEA optimization problem is

$$\min \sum_{t \in \mathcal{T}} \sum_{n \in \mathcal{N}_c} \left(\lambda_{n,t}^p \cdot p_{n,t}^c + \lambda_{n,t}^q \cdot q_{n,t}^c + \pi \cdot (\hat{P}_{n,t}^{\text{load}} - p_{n,t}^{\text{load}}) \right) \quad (2)$$

$$\text{subject to: } 0 \leq p_{n,t}^{\text{pv}} \leq \hat{P}_{n,t}^{\text{pv}} \quad (3a)$$

$$q_{n,t}^{\text{pv}} \leq p_{n,t}^{\text{pv}} \zeta^{\text{pv}} \quad (3b)$$

$$-q_{n,t}^{\text{pv}} \leq p_{n,t}^{\text{pv}} \zeta^{\text{pv}} \quad (3c)$$

$$0 \leq p_{n,t}^{\text{ch}} \leq \bar{P}_n^{\text{ch}} \cdot b_{n,t}^{\text{ch}} \quad (3d)$$

$$0 \leq p_{n,t}^{\text{dis}} \leq \bar{P}_n^{\text{dis}} \cdot b_{n,t}^{\text{dis}} \quad (3e)$$

$$b_{n,t}^{\text{ch}} + b_{n,t}^{\text{dis}} \leq 1, \quad b_{n,t}^{\text{ch}}, b_{n,t}^{\text{dis}} \in \{0, 1\} \quad (3f)$$

$$e_{n,t+1} = e_{n,t} + \eta_n^{\text{ch}} p_{n,t}^{\text{ch}} \Delta t - (1/\eta_n^{\text{dis}}) p_{n,t}^{\text{dis}} \Delta t \quad (3g)$$

$$\underline{E}_n \leq e_{n,t} \leq \bar{E}_n \quad (3h)$$

$$0 \leq p_{n,t}^{\text{load}} \leq \hat{P}_{n,t}^{\text{load}} \quad (3i)$$

$$p_{n,t}^c = p_{n,t}^{\text{load}} - p_{n,t}^{\text{pv}} - p_{n,t}^{\text{dis}} + p_{n,t}^{\text{ch}} \quad (3j)$$

$$q_{n,t}^c = \hat{Q}_{n,t}^{\text{load}} - q_{n,t}^{\text{pv}} \quad (3k)$$

Here, $\lambda_{n,t}^p$ and $\lambda_{n,t}^q$ represent DLMP variables for active and reactive power and are treated as input parameters in the CEA problem. Note that the reactive power $q_{n,t}^c$ can be either positive or negative, so $\lambda_{n,t}^q q_{n,t}^c$ represents either a cost for net reactive consumption or a credit for reactive support, depending on the sign of $q_{n,t}^c$. Similar active/reactive local pricing and reactive-power market mechanisms have been considered in recent distribution-level market studies [27], [28]. They are, however, determined as dual variables associated with power balance constraints in the DSO model described in Section II-C. It is important to note that the CEA only considers community buses in its objective function and excludes the operation of other non-community buses.

The CEA scheduling problem includes constraints on DER operations across all community buses $n \in \mathcal{N}_c$ and time periods $t \in \mathcal{T}$. These constraints consist of (i) PV generation limits, (ii) BESS operating rules, (iii) controllable load bounds, and (iv) the resulting net power consumptions submitted to the DSO. The complete set of constraints is outlined as follows.

The PV operating limits are defined by (3a)–(3c), including real power availability and a linear approximation of the inverter reactive-power capability. The BESS model, given

in (3d)–(3h), captures charging and discharging limits, state-of-charge dynamics, and mutually exclusive operating modes. Controllable loads are bounded within their flexibility limits as specified in (3i). The net active and reactive power consumption at each community bus is determined by (3j) and (3k). The net injection reflects the scheduled load minus local DER outputs, including PV generation and BESS discharging (or charging). These net injections are submitted to the DSO for network-constrained dispatch and DLMP calculation. The BESS units are assumed not to provide reactive power support.

C. DSO Model: Lower Level Problem

The DSO aims to ensure the reliable and cost-effective operation of the distribution grid in response to the power injections scheduled by the CEA. Specifically, it minimizes the total procurement and dispatch costs while maintaining system-wide power balance and voltage limits. The upstream power purchase is settled at the wholesale market price (π_t^{LMP}), while dispatchable local generators are operated based on their marginal costs. In this model, the DSO determines the DLMPs through the dual variables of the power balance constraints, which are used by the CEA as price signals for subsequent scheduling. The DSO optimization problem is

$$\min \sum_{t \in \mathcal{T}} \pi_t^{\text{LMP}} \cdot p_{0,t}^g + \sum_{t \in \mathcal{T}} \sum_{n \in \mathcal{N} \setminus \mathcal{N}_c} c_n^g \cdot p_{n,t}^g \quad (4)$$

subject to:

$$u_{j,t} = u_{i,t} - 2(r_{ij} P_{ij,t} + x_{ij} Q_{ij,t}) : \lambda_{ij,t}^{\text{volt}} \quad (5a)$$

$$(v^{\min})^2 \leq u_{n,t} \leq (v^{\max})^2 : \underline{\lambda}_{n,t}^u, \bar{\lambda}_{n,t}^u \quad (5b)$$

$$\sum_{j: (0,j) \in \mathcal{L}} P_{0j,t} - p_{0,t}^{\text{inj}} - p_{0,t}^g = 0 : \lambda_{0,t}^p \quad (5c)$$

$$\sum_{j: (n,j) \in \mathcal{L}} P_{nj,t} - \sum_{i: (i,n) \in \mathcal{L}} P_{in,t} - p_{n,t}^{\text{inj}} - p_{n,t}^g = 0 : \lambda_{n,t}^p \quad (5d)$$

$$\sum_{j: (0,j) \in \mathcal{L}} Q_{0j,t} - q_{0,t}^{\text{inj}} - q_{0,t}^g = 0 : \lambda_{0,t}^q \quad (5e)$$

$$\sum_{j: (n,j) \in \mathcal{L}} Q_{nj,t} - \sum_{i: (i,n) \in \mathcal{L}} Q_{in,t} - q_{n,t}^{\text{inj}} - q_{n,t}^g = 0 : \lambda_{n,t}^q \quad (5f)$$

$$\underline{p}_n^g \leq p_{n,t}^g \leq \bar{p}_n^g : \underline{\lambda}_{n,t}^{gp}, \bar{\lambda}_{n,t}^{gp} \quad (5g)$$

$$\underline{q}_n^g \leq q_{n,t}^g \leq \bar{q}_n^g : \underline{\lambda}_{n,t}^{gq}, \bar{\lambda}_{n,t}^{gq} \quad (5h)$$

Here, (5a)–(5f) represent the power-flow model using the LinDistFlow equations, ensuring a tractable convex formulation [15]. In this approximation, quadratic power loss terms are neglected. Constraints (5g) and (5h) limit the generators' active and reactive power outputs. Note that these constraints are defined for all buses $n \in \mathcal{N}$, lines $(i,j) \in \mathcal{L}$, and time periods $t \in \mathcal{T}$, although such indices are omitted for brevity. The resulting linearized constraints and their associated Lagrange multipliers $\lambda^\bullet$ are shown next to the constraints. In the power balance constraints (5d) and (5f), power injections are related to the net active and reactive power consumption of the connected community and load forecasts as $p_{n,t}^{\text{inj}} = -p_{n,t}^c, \forall n \in \mathcal{N}_c$ otherwise $p_{n,t}^{\text{inj}} = -\hat{P}_{n,t}^{\text{load}}, \forall n \in \mathcal{N} \setminus \mathcal{N}_c$. Similarly, reactive power injections follow $q_{n,t}^{\text{inj}} = -q_{n,t}^c, \forall n \in \mathcal{N}_c$ and $q_{n,t}^{\text{inj}} = -\hat{Q}_{n,t}^{\text{load}}, \forall n \in \mathcal{N} \setminus \mathcal{N}_c$.

D. Single-level Reformulation via KKT Conditions

As described in Section II-A, the interaction between the CEA and the DSO is formulated as a bilevel optimization problem, where the CEA (upper-level) determines community-level energy schedules, and the DSO (lower-level) responds with optimal dispatch and pricing decisions. To obtain a tractable formulation, the lower-level DSO problem is replaced by its KKT optimality conditions [29]. We start by writing the Lagrangian using the DSO objective and constraints (5) as

$$\begin{aligned}
\mathcal{L} = & \sum_{t \in \mathcal{T}} \pi_t^{\text{LMP}} \cdot p_{0,t}^g + \sum_{t \in \mathcal{T}} \sum_{n \in \mathcal{N} \setminus \mathcal{N}_s} c_n^g \cdot p_{n,t}^g \\
& + \sum_{t \in \mathcal{T}} \sum_{(i,j) \in \mathcal{L}} \lambda_{ij,t}^{\text{volt}} (u_{j,t} - u_{i,t} + 2(r_{ij} P_{ij,t} + x_{ij} Q_{ij,t})) \\
& + \sum_{t \in \mathcal{T}} \sum_{n \in \mathcal{N}} \left[\lambda_{n,t}^u ((v^{\min})^2 - u_{n,t}) + \bar{\lambda}_{n,t}^u (u_{n,t} - (v^{\max})^2) \right] \\
& + \sum_{t \in \mathcal{T}} \lambda_{0,t}^p \left(\sum_{j: (0,j) \in \mathcal{L}} P_{0j,t} - p_{0,t}^{\text{inj}} - p_{0,t}^g \right) \\
& + \sum_{t \in \mathcal{T}} \sum_{n \in \mathcal{N} \setminus \mathcal{N}_s} \lambda_{n,t}^p \left(\sum_{j: (n,j) \in \mathcal{L}} P_{nj,t} - \sum_{i: (i,n) \in \mathcal{L}} P_{in,t} - p_{n,t}^{\text{inj}} - p_{n,t}^g \right) \\
& + \sum_{t \in \mathcal{T}} \lambda_{0,t}^q \left(\sum_{j: (0,j) \in \mathcal{L}} Q_{0j,t} - q_{0,t}^{\text{inj}} - q_{0,t}^g \right) \\
& + \sum_{t \in \mathcal{T}} \sum_{n \in \mathcal{N} \setminus \mathcal{N}_s} \lambda_{n,t}^q \left(\sum_{j: (n,j) \in \mathcal{L}} Q_{nj,t} - \sum_{i: (i,n) \in \mathcal{L}} Q_{in,t} - q_{n,t}^{\text{inj}} - q_{n,t}^g \right) \\
& + \sum_{t \in \mathcal{T}} \sum_{n \in \mathcal{N}} \left[\lambda_{n,t}^{gp} (p_n^g - p_{n,t}^g) + \bar{\lambda}_{n,t}^{gp} (p_{n,t}^g - \bar{p}_n^g) \right] \\
& + \sum_{t \in \mathcal{T}} \sum_{n \in \mathcal{N}} \left[\lambda_{n,t}^{gq} (q_n^g - q_{n,t}^g) + \bar{\lambda}_{n,t}^{gq} (q_{n,t}^g - \bar{q}_n^g) \right] \quad (6)
\end{aligned}$$

Now, we can write the DSO problem's KKT conditions, consisting of four components: primal feasibility, stationarity, dual feasibility, and complementarity. Primal feasibility requires satisfying all DSO constraints, including the LinDistFlow equations and operational limits defined in (5).

Let $x_{\text{DSO}} := \{P_{ij,t}, Q_{ij,t}, u_{n,t}, p_{n,t}^g, q_{n,t}^g\}$ denote the set of DSO primal variables for $(i,j) \in \mathcal{L}$, $n \in \mathcal{N}$, and $t \in \mathcal{T}$. Stationarity requires that the gradient of the Lagrangian with respect to each element of x_{DSO} vanishes:

$$\frac{\partial \mathcal{L}}{\partial P_{ij,t}} = 2r_{ij} \lambda_{ij,t}^{\text{volt}} + \lambda_{i,t}^p - \lambda_{j,t}^p = 0 \quad (7a)$$

$$\frac{\partial \mathcal{L}}{\partial Q_{ij,t}} = 2x_{ij} \lambda_{ij,t}^{\text{volt}} + \lambda_{i,t}^q - \lambda_{j,t}^q = 0 \quad (7b)$$

$$\frac{\partial \mathcal{L}}{\partial u_{n,t}} = - \sum_{j: (n,j) \in \mathcal{L}} \lambda_{nj,t}^{\text{volt}} + \sum_{i: (i,n) \in \mathcal{L}} \lambda_{in,t}^{\text{volt}} - \lambda_{n,t}^u + \bar{\lambda}_{n,t}^u = 0 \quad (7c)$$

$$\frac{\partial \mathcal{L}}{\partial p_{0,t}^g} = \pi_t^{\text{LMP}} - \lambda_{0,t}^p - \lambda_{0,t}^{gp} + \bar{\lambda}_{0,t}^{gp} = 0 \quad (7d)$$

$$\frac{\partial \mathcal{L}}{\partial q_{0,t}^g} = -\lambda_{0,t}^q - \lambda_{0,t}^{gq} + \bar{\lambda}_{0,t}^{gq} = 0 \quad (7e)$$

$$\frac{\partial \mathcal{L}}{\partial p_{n,t}^g} = c_n^g - \lambda_{n,t}^p - \lambda_{n,t}^{gp} + \bar{\lambda}_{n,t}^{gp} = 0 \quad (7f)$$

$$\frac{\partial \mathcal{L}}{\partial q_{n,t}^g} = -\lambda_{n,t}^q - \lambda_{n,t}^{gq} + \bar{\lambda}_{n,t}^{gq} = 0. \quad (7g)$$

Let $\mathcal{I}$ denote the index set of all DSO inequality constraints, including voltage magnitude limits and active/reactive generation capacity limits, each written as $g_i(x_{\text{DSO}}) \leq 0$ with associated dual variable λ_i . Dual feasibility requires

$$\lambda_i \geq 0, \quad \forall i \in \mathcal{I}. \quad (8)$$

Complementarity conditions between each inequality constraint and its corresponding dual variable are enforced using a Big- M reformulation [30], where $\mathfrak{M}$ denotes a sufficiently large positive constant. For each $i \in \mathcal{I}$, a binary variable $\delta_i \in \{0, 1\}$ is introduced, yielding

$$0 \leq \lambda_i \leq \mathfrak{M} \delta_i, \quad -\mathfrak{M}(1 - \delta_i) \leq g_i(x_{\text{DSO}}) \leq 0, \quad \forall i \in \mathcal{I}. \quad (9)$$

The original CEA objective (2) contains the DLMP-based terms $\lambda_{n,t}^p p_{n,t}^c$ and $\lambda_{n,t}^q q_{n,t}^c$. After replacing the lower-level DSO problem by its KKT conditions, the multipliers $\lambda_{n,t}^p$ and $\lambda_{n,t}^q$ become decision variables of the single-level problem, which makes it bilinear in (λ, p^c, q^c) . Since the DSO problem is linear and feasible, strong duality holds. Thus, the primal and dual DSO objectives are equal. Using this property and as shown in Appendix A, the bilinear terms in (2) are replaced by their dual representation, yielding the linear objective in (10).

$$\begin{aligned}
f^{\text{SINGLE}} = & \sum_{t \in \mathcal{T}} \pi_t^{\text{LMP}} p_{0,t}^g + \sum_{t \in \mathcal{T}} \sum_{n \in \mathcal{N} \setminus \mathcal{N}_s} c_n^g p_{n,t}^g \\
& - \sum_{t \in \mathcal{T}} \sum_{n \in \mathcal{N} \setminus \mathcal{N}_c} \lambda_{n,t}^p \hat{P}_{n,t}^{\text{load}} - \sum_{t \in \mathcal{T}} \sum_{n \in \mathcal{N} \setminus \mathcal{N}_c} \lambda_{n,t}^q \hat{Q}_{n,t}^{\text{load}} \\
& + \sum_{t \in \mathcal{T}} \sum_{n \in \mathcal{N}_c} \pi (\hat{P}_{n,t}^{\text{load}} - p_{n,t}^{\text{load}}) \\
& + \sum_{t \in \mathcal{T}} \sum_{n \in \mathcal{N}} \left(\bar{\lambda}_{n,t}^{gp} \bar{p}_n^g - \lambda_{n,t}^{gp} p_n^g \right) + \sum_{t \in \mathcal{T}} \sum_{n \in \mathcal{N}} \left(\bar{\lambda}_{n,t}^{gq} \bar{q}_n^g - \lambda_{n,t}^{gq} q_n^g \right) \\
& + \sum_{t \in \mathcal{T}} \sum_{n \in \mathcal{N}} \left(\bar{\lambda}_{n,t}^u (v^{\max})^2 - \lambda_{n,t}^u (v^{\min})^2 \right). \quad (10)
\end{aligned}$$

Note that all terms in the objective function are linear as a result of the dual substitution. Finally, the original bilevel CEA-DSO problem is reformulated as the following single-level mixed-integer linear program (MILP):

$$\begin{aligned}
\min \quad & f^{\text{SINGLE}} \\
\text{subject to:} \quad & \text{CEA constraints: (3),} \\
& \text{DSO KKT conditions: (5), (7), (8), (9)}
\end{aligned} \quad (11)$$

This reformulation removes the nested structure of the bilevel problem by embedding the DSO optimality conditions into the upper-level model, as illustrated in Fig. 2.

III. FAIR COST ALLOCATION USING COALITIONAL GAME AND SHAPLEY VALUE METHOD

The optimal value obtained by (11) optimizes the total operational cost aggregated over all participating ECs. Although this cost is optimal from the CEA's perspective, it does not necessarily ensure fairness in the allocation of costs among individual ECs. This motivates the use of a formal fairness-based allocation mechanism. Without a mechanism that reflects each EC's contribution to cooperative cost savings, the resulting allocation may discourage ECs from providing flexibility or investing in grid-supportive DERs.

In this section, we present a fair allocation approach based on the Shapley value, which formulates the allocation of cooperative savings as a coalitional game and distributes the savings according to each EC's marginal contribution. Since the practical question addressed in this paper is the allocation of realized cooperative cost savings among individual ECs, we adopt this fairness-oriented approach rather than stability-oriented concepts such as the core, which is the set of allocations under which no coalition has an incentive to deviate from the grand coalition, or its relaxation, the least core, which seeks an allocation that minimizes the maximum incentive for any coalition to deviate from the grand coalition [31]. It should be noted that this Shapley-based allocation does not replace or average DLMPs across buses; instead, DLMPs are retained as operational price signals, while the Shapley value is used as an ex-post settlement rule for allocating cooperative savings.

Let $\mathcal{M}$ denote the set of participating ECs, and let $\mathcal{C} \subseteq \mathcal{M}$ represent a coalition. To connect the bus-indexed optimization framework in Section II with the community-indexed coalitional game, let $b(m) \in \mathcal{N}_c$ denote the bus associated with EC m . The bus-indexed variables $p_{b(m),t}^c$ and $q_{b(m),t}^c$ then represent the active and reactive power consumption of EC m , and a coalition $\mathcal{C} \subseteq \mathcal{M}$ is associated with the set of community buses $\{b(m) : m \in \mathcal{C}\}$. The characteristic function $v : 2^{\mathcal{M}} \rightarrow \mathbb{R}$ assigns to each coalition $\mathcal{C}$ the corresponding cooperative cost savings. The individual cost c_m^{indiv} of each EC $m \in \mathcal{M}$ is obtained by solving (11) with only EC m participating, while all other ECs operate under their baseline profiles. For ECs outside the evaluated coalition, the baseline profiles correspond to uncontrolled operation: PV output follows the forecasted available generation, BESS operation is disabled, and load reduction is not allowed.

Similarly, for a coalition $\mathcal{C} \subseteq \mathcal{M}$, the cooperative cost $c_{\mathcal{C}}$ is computed by jointly optimizing the ECs in $\mathcal{C}$, whereas the ECs outside the coalition, i.e., those in $\mathcal{M} \setminus \mathcal{C}$, remain under their baseline profiles. Here, $c_{\mathcal{C}}$ denotes the optimal aggregate operating cost of the ECs in coalition $\mathcal{C}$ obtained from the single-level CEA-DSO problem (11). The benefit redistributed among the participating ECs is the cooperative cost saving achieved through their joint scheduling. Accordingly, the value of coalition $\mathcal{C}$ is defined as the total cost saving relative to individual operation as:

$$v(\mathcal{C}) = \sum_{m \in \mathcal{C}} c_m^{\text{indiv}} - c_{\mathcal{C}}. \quad (12)$$

Using the Shapley value method [7], the allocation $\phi_m(v)$ to community m for a set of communities $\mathcal{M}$ with $|\mathcal{M}| = N$, is defined as:

$$\phi_m(v) = \sum_{\mathcal{S} \subseteq \mathcal{M} \setminus \{m\}} \frac{|\mathcal{S}|!(N - |\mathcal{S}| - 1)!}{N!} [v(\mathcal{S} \cup \{m\}) - v(\mathcal{S})], \quad (13)$$

where $v(\mathcal{S} \cup \{m\}) - v(\mathcal{S})$ represents the marginal contribution of community m to a coalition $\mathcal{S}$. This allocation satisfies the efficiency property, $\sum_{m \in \mathcal{M}} \phi_m(v) = v(\mathcal{M})$, ensuring that the total savings from joint optimization are fully and fairly distributed based on each community's contribution.

The final cost for each community is obtained by subtracting its share of the cost savings from its individual cost:

$$c_m^{\text{final}} = c_m^{\text{indiv}} - \phi_m(v) \quad (14)$$

Note that the optimization problems associated with different coalitions are independent and can therefore be solved in parallel, enabling efficient computation for Shapley value evaluation. However, despite parallelization, the total number of coalitions grows exponentially with the number of ECs, limiting the scalability of full enumeration for large systems. We mitigate this issue by proposing a signature-based approximation method that efficiently selects a representative subset of coalitions for evaluation and reconstructs the remaining coalition values. The overall procedure is summarized in Algorithm 1, while the screening criteria and computational performance are presented in Section IV.

Algorithm 1 Signature-Based Shapley Reconstruction

Require: EC set $\mathcal{M}$, characteristic function $v(\cdot)$, network topology

Ensure: Approximate Shapley allocation $\hat{\phi}_m, \forall m \in \mathcal{M}$

Step 1: Collect screening information

- 1: Evaluate $v(\{i\})$ and $v(\{i, j\})$ for all $i, j \in \mathcal{M}$.
- 2: Extract singlet-EC values, pairwise marginal effects, and feeder-section information.

Step 2: Construct representative coalitions

- 3: Group ECs into representative groups based on the screening information collected in Step 1.
- 4: Generate the representative coalition set $\mathcal{R} \subseteq 2^{\mathcal{M}}$.
- 5: Evaluate $v(\hat{S})$ only for representative coalitions $\hat{S} \in \mathcal{R}$.
- 6: Define the mapping $\rho : 2^{\mathcal{M}} \rightarrow \mathcal{R}$ that assigns each coalition $S \subseteq \mathcal{M}$ to its representative coalition $\rho(S)$.

Step 3: Reconstruct Shapley values

- 7: **for** each coalition $S \subseteq \mathcal{M}$ **do**
 - 8: Set $\hat{v}(S) = v(\rho(S))$.
 - 9: **end for**
 - 10: Compute $\hat{\phi}_m$ using the Shapley formula in (13).
 - 11: **return** approximate Shapley allocations $\hat{\phi}_m$.
-

IV. NUMERICAL VALIDATION

A. Simulation Setup

We evaluate the proposed fair allocation scheme on four benchmark distribution systems: the CIGRE low-voltage [32], the IEEE 69-bus [15], the IEEE 123-bus [33], and the 9500-bus [34] systems. First, the CIGRE system is used to illustrate the proposed allocation framework and its comparison with existing methods, while the other test-cases are used to evaluate its computational performance, scalability, and robustness.

Across all test cases, the same modeling assumptions and DER parameters are applied. Each EC is composed of PV plants, BESS, and controllable loads. We assume the loads can be controlled up to 30% of their base demand through flexible load response, with a compensation rate of \$75/MWh [35]. The electricity price at the substation uses the 2024 average of hourly CAISO locational marginal prices (LMP) for the NP-15 zone, which varies between \$12/MWh and \$55/MWh [36]. Local dispatchable generators are assumed to be available

at a few distribution buses, operating at a marginal cost of \$250/MWh, based on values reported in [37]. The reactive power capabilities of dispatchable generators and PV inverters are constrained by typical grid code requirements. Generator output is limited by a minimum power factor of 0.9, while PV inverters are restricted to supplying reactive power up to 20% of their real power output [27]. The charging and discharging efficiencies of BESS are both set to 95%, i.e., $\eta_n^{\text{ch}} = \eta_n^{\text{dis}} = 0.95$. The initial and final state of charge (SoC) is set to 50% of the rated capacity. We model the load and PV generation using real measurements from the smart grid setup at the EPFL, Switzerland [38]. The data,¹ which is available at per second resolution, is down-sampled to hourly resolution.

For each EC, we show four quantities: (i) *individual cost* c_m^{indiv} is obtained by solving (11) with only one particular EC active while others are uncontrolled and uncoordinated. This represents the cost if each EC operates independently without coordination, (ii) The *Shapley-based saving* $\phi_m(v)$ quantifies the marginal contribution of community m to the grand coalition (all ECs are jointly optimized for their coordinated operation), (iii) the *Final cost (Shapley)* settlement for each community is calculated as $c_m^{\text{final}} = c_m^{\text{indiv}} - \phi_m(v)$ according to (14), ensuring that the cost reduction from cooperation is allocated based on the contribution of each participant, and finally (iv) the *Final cost (Base)* cost, c_m^{base} where the cost is directly calculated from the DLMPs and dispatch from the solution to (11), without accounting for the community's marginal cost savings. We use YALMIP [39] to implement the optimization problem, and Gurobi as solver. Simulations are done in MATLAB on a laptop with 24 GB of memory.

B. Simulation Results

1) CIGRE 19-bus system

Fig. 3 shows the benchmark CIGRE low-voltage distribution network used in this study. The feeder is connected to the transmission system via a 20/0.4 kV, 400 kVA transformer. The system includes both community and non-community loads, and the impact of community locations on system-level fairness and grid operation is analyzed for multiple scenarios.

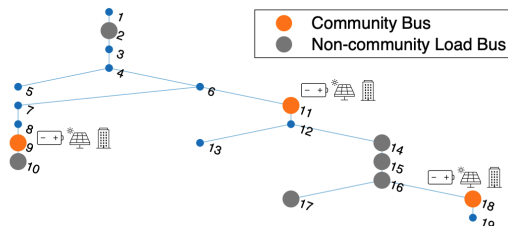


Fig. 3. CIGRE low voltage benchmark network. Local dispatchable generators are assumed to be available at buses 13, 16, and 18.

In the first scenario, we consider three communities located at buses 9, 11, and 18. Each community is initially assumed to have an identical DER setup consisting of PV, controllable demand, and a BESS rated at 20 kW/50 kWh. Fig. 4 compares the individual costs, the Shapley-based savings, and the final allocated costs under the Shapley and Base methods. We observe that communities under cooperation, i.e., in Final cost with Shapley and Base, the costs are lower compared to the

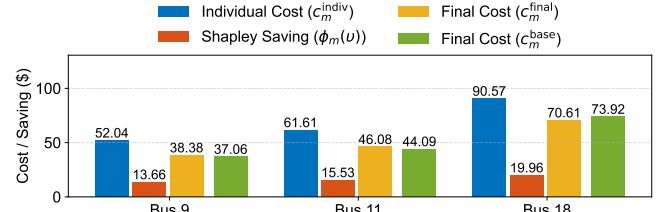


Fig. 4. Comparison of individual cost, Shapley-based savings, and final allocated cost under the Shapley and Base methods for communities at buses 9, 11, and 18.

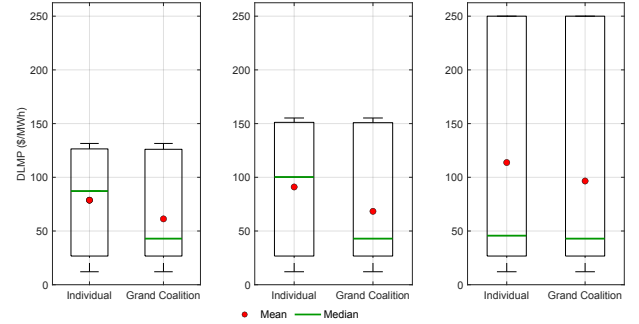


Fig. 5. Active power DLMP distributions at the community buses under individual participation and grand coalition participation. In the individual case, only one EC is optimized while the others follow baseline (uncontrolled) profiles; in the grand coalition case, all ECs in $\mathcal{M}$ are jointly optimized.

without any cooperation (individual). Moreover, the Shapley method distributes the savings fairly based on marginal contributions, whereas the Base method directly uses DLMP results without considering individual contributions, which is lower than individual costs, but may be unfair.

Among the three, the EC located at bus 18 exhibits the highest individual and final costs, along with the largest Shapley value. This can be explained by distributions of DLMPs for individual and cooperative case, shown in Fig. 5. As shown, the DLMPs at bus 18 are significantly higher, which might be due to binding voltage constraints. As a result, its contribution to overall system cost reduction is the most significant, and is reflected in its higher Shapley value. Across all buses, DLMP distributions shift lower under the grand coalition, indicating that coordinated participation benefits all communities.

a) *Sensitivity with BESS size:* To study the impact of BESS sizes on DLMPs and fairness, we consider scenarios of different BESS capacities at a single site. An additional BESS enhances operational flexibility and enables peak shaving, leading to reduced system-wide costs. However, how these savings are distributed disproportionately across communities depending on the allocation rule.

Table II presents the final Shapley and Base cost of each community when the BESS at one site is increased to three times its original capacity. The Shapley allocation more accurately attributes cost savings to the community providing additional BESS capacity. When bus 11 hosts the expanded BESS, its cost decreases from \$46.08 to \$26.28 (a \$19.80 reduction). In contrast, under the Base allocation, bus 11's cost decreases by only \$11.29 (from \$44.09 to \$32.80), while bus 18, despite contributing no additional resources, also benefits from a \$8.01 reduction. This illustrates that the baseline method unfairly benefits non-contributing communities and does not fully capture localized contributions.

Table III further extends the analysis by expanding BESS

¹<https://github.com/DESL-EPFL/DESL-Photovoltaic-timeseries>

TABLE II

FINAL COST COMPARISON UNDER BESS EXPANSION ($3 \times$ ORIGINAL CAPACITY) AT A SINGLE COMMUNITY. "COMM." DENOTES THE BUS ID OF EACH COMMUNITY.

Method	Scenario	Comm.	BESS x1 Cost (\$)	BESS x3 Cost (\$)	Cost Diff. (\$)
Shapley	BESS@9 x3	Bus 9	38.38	20.90	17.48
		Bus 11	46.08	44.37	1.71
		Bus 18	70.61	68.31	2.30
	Total		155.07	133.58	21.49
	BESS@11 x3	Bus 9	38.38	36.99	1.39
		Bus 11	46.08	26.28	19.80
		Bus 18	70.61	68.91	1.70
	Total		155.07	132.19	22.89
	BESS@18 x3	Bus 9	38.38	38.81	-0.43
		Bus 11	46.08	43.24	2.84
		Bus 18	70.61	39.30	31.31
	Total		155.07	121.36	33.72
Base	BESS@9 x3	Bus 9	37.06	26.44	10.62
		Bus 11	44.09	41.09	3.00
		Bus 18	73.92	66.06	7.87
	Total		155.07	133.58	21.49
	BESS@11 x3	Bus 9	37.06	33.47	3.59
		Bus 11	44.09	32.80	11.29
		Bus 18	73.92	65.91	8.01
	Total		155.07	132.19	22.89
	BESS@18 x3	Bus 9	37.06	29.09	7.96
		Bus 11	44.09	35.96	8.14
		Bus 18	73.92	56.31	17.62
	Total		155.07	121.36	33.72

TABLE III

FINAL COST COMPARISON WHEN EXPANDING BESS AT BUS 11 TO $3.5 \times$ ITS ORIGINAL CAPACITY.

Method	Scenario	Comm.	BESS x1 Cost (\$)	BESS x3.5 Cost (\$)	Cost Diff. (\$)
Shapley	BESS@11	Bus 9	38.38	35.30	3.08
		Bus 11	46.08	20.90	25.18
		Bus 18	70.61	67.76	2.86
	Total		155.07	123.95	31.12
Base	BESS@11	Bus 9	37.06	30.67	6.38
		Bus 11	44.09	32.85	11.24
		Bus 18	73.92	60.42	13.50
	Total		155.07	123.95	31.12

capacity at bus 11 to 3.5 times its original size. Under the Shapley method, the total cost decreases from \$155.07 to \$123.95, with the largest benefit concentrated at bus 11, whose cost drops from \$46.08 to \$20.90 (\$25.18 reduction). In contrast, under the **Base** method, the cost at bus 11 decreases from \$44.09 to \$32.85 (\$11.24 reduction), while bus 18, without any new investment, sees its cost drop from \$73.92 to \$60.42 (\$13.50 reduction), which is even larger. This result highlights a potential limitation of the **Base** allocation: it may not fully reflect the origin of the cost savings, which could result in a less fair allocation. Such misalignment between contribution and benefit may weaken the incentive for communities to invest in grid-supportive resources like BESS.

In one of the cases, we encounter negative cost difference (bus 9 under the "BESS@18 x3" scenario) where the allocated cost under the Shapley method is slightly higher than the original cost. However, the increase is marginal, and the cost

TABLE IV

TOTAL SYSTEM DISPATCH COST BY THE NUMBER OF PARTICIPATING ECS IN THE CIGRE 19-BUS SYSTEM.

# Participating ECs	Avg. cost (\$)	Min. cost (\$)	Max. cost (\$)
0	356.16	356.16	356.16
3	286.38	278.46	300.50
5	257.67	257.67	257.67

still remains lower than the individual cost (Fig. 4). Therefore, cooperation is still beneficial for all communities, and the individual rationality condition is satisfied.

b) *Total system dispatch cost under coalition operation:*

We next examine how EC coalition operation affects system-level cost. The participating ECs are located at buses 3, 7, 9, 12, and 18, while buses 2, 10, 14, 15, 16, and 17 serve as non-community load buses. Table IV reports the average total system dispatch cost, measured by the DSO objective value in (4), for selected coalition sizes. The average cost decreases from \$356.16 under the uncoordinated baseline to \$257.67 under the grand coalition, corresponding to a 27.65% reduction. This reduction occurs because participating ECs provide additional operational flexibility through DER scheduling, which helps reduce upstream procurement and local dispatch costs. We also observe that the average system dispatch cost decreases with the number of participating ECs, indicating that broader coalition participation tends to improve system-level operational efficiency in this case.

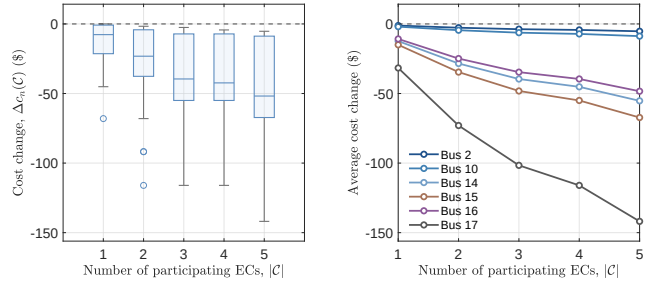


Fig. 6. Cost impacts of EC coalition operation on non-community load buses. The left panel shows the distribution of $\Delta c_n(\mathcal{C})$, and the right panel shows the bus-level average cost changes on non-community load buses.

c) *DLMP-based cost impacts on non-community load buses:*

Under the same setting, we next analyze whether EC coalition operation adversely affects users outside the coalition. For each coalition $\mathcal{C} \subseteq \mathcal{M}$ of participating ECs, we compute the 24-hour aggregate DLMP-based cost at each non-community load bus n as

$$c_n(\mathcal{C}) = - \sum_{t \in \mathcal{T}} \left(\lambda_{n,t}^p(\mathcal{C}) p_{n,t}^{\text{inj}}(\mathcal{C}) + \lambda_{n,t}^q(\mathcal{C}) q_{n,t}^{\text{inj}}(\mathcal{C}) \right), \quad (15)$$

where $\lambda_{n,t}^p(\mathcal{C})$ and $\lambda_{n,t}^q(\mathcal{C})$ are the active and reactive DLMPs at bus n and time t under coalition $\mathcal{C}$, respectively. We then compare this cost with the no-coordination baseline as

$$\Delta c_n(\mathcal{C}) = c_n(\mathcal{C}) - c_n(\emptyset). \quad (16)$$

Thus, negative values of $\Delta c_n(\mathcal{C})$ indicate cost reductions for non-community load buses, whereas positive values indicate cost increases. Fig. 6 reports the cost impacts as the coalition size increases. For each coalition size $|\mathcal{C}|$, the left panel shows the distribution of $\Delta c_n(\mathcal{C})$ across all coalition configurations

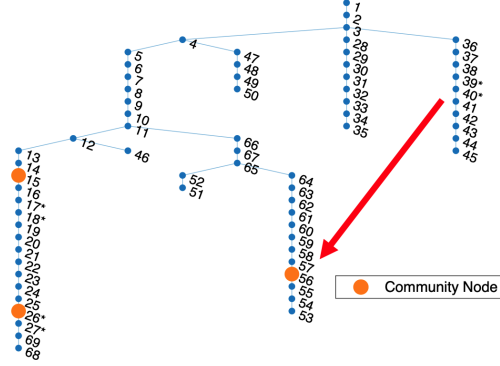


Fig. 7. IEEE 69-bus distribution system showing three different EC placement configurations. The first two configurations consider three ECs located at buses 15, 26, and 40 and at buses 15, 26, and 56, respectively. The third configuration places six ECs at buses 17, 18, 26, 27, 39, and 40, marked with an asterisk (*) to demonstrate the signature-based approximation.

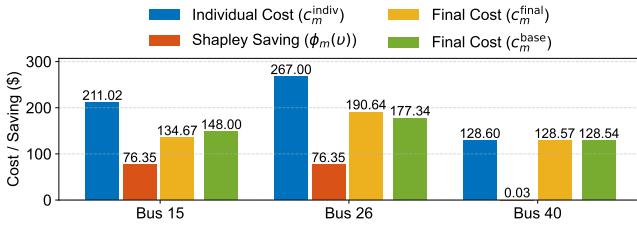


Fig. 8. Individual cost, Shapley-based savings, and final cost for each community bus in the IEEE 69-bus system with ECs at buses 15, 26, and 40, where the EC at bus 40 has limited interaction with the others.

and non-community load buses (i.e., buses 2, 10, 14, 15, 16, and 17). The right panel shows the average cost change at each non-community load bus. The results demonstrate that the average cost of the non-participating load buses decreases as more ECs join the coalition, indicating that coordinated EC operation can also reduce DLMP-based costs for users outside the coalition. However, the spread of the box plots and the bus-dependent trends also indicate that the magnitude of these benefits depends on both network location and coalition composition. Therefore, although no adverse impacts are observed in the tested cases, practical settlement frameworks should monitor cost changes at non-community load buses to ensure that coalition savings are not achieved at their expense.

2) IEEE 69-bus system

We extend the analysis to the IEEE 69-bus test system, with the EC placements shown in Fig. 7. We consider three scenarios with different EC placements to examine the impact of community location on cost savings. All ECs are assumed to have identical DER portfolios and capacities. We compare Shapley and Base methods as defined earlier.

a) *Case 1 - ECs at buses 15, 26 and 40:* We first examine a configuration where one EC is placed farther away (i.e., bus 40) from the others (i.e., buses 15 and 26). The results are visualized in Fig. 8. As observed, the cost for EC at bus 40 changes only slightly through cooperation. This may be due to weak coupling between the EC's operations as they are far apart. In contrast, the ECs at buses 15 and 26 exhibit equal Shapley savings, reflecting their symmetric contributions to system-wide cost savings. This symmetry is not seen under the Base method, which leads to a relatively more favorable allocation for bus 26, illustrating the base method's inability to

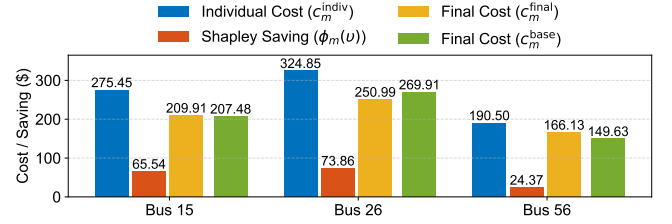


Fig. 9. Individual cost, Shapley-based savings, and final cost for each community bus in the IEEE 69-bus system with ECs at buses 15, 26, and 56, where all three ECs interact more strongly.

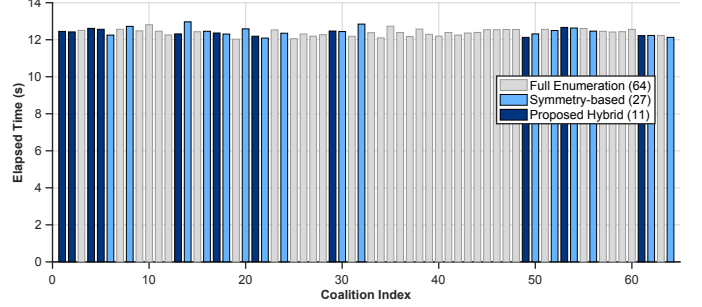


Fig. 10. Elapsed time for all 64 coalitions (Full Enumeration). The representative coalitions correspond to the 27 symmetry-based (light blue) and the 11 proposed hybrid (navy blue) scenarios.

fully reflect contribution-based fairness. Nevertheless, the contributing ECs still benefit from participating in the coalition, with final costs lower than their individual operation costs, preserving the principle of individual rationality.

b) *Case 2 - ECs at buses 15, 26 and 56:* Next, we relocate the EC at bus 40 to bus 56 to study the impact of location on the cost saving. In this case, the computed cost savings are non-zero for all three ECs, as shown in Fig. 9, which reflects that each EC contributes non-negligibly to the coalition's performance. The result above highlights that the Shapley allocation captures the locational impacts of DER placements and ensures a fair cost distribution.

c) *Scalability with more ECs:* To investigate computational scalability, we expand the study to six ECs at buses 17, 18, 26, 27, 39, and 40, resulting in $2^6 = 64$ coalitions. Evaluating the costs for all the coalitions takes substantial computation time, therefore we propose a “signature-based approximation” which aims to reduce the number of coalitions to be evaluated to obtain same results as when all the coalitions are evaluated.

The proposed approximation leverages two physical insights. *First*, we cluster the ECs with identical DER configurations and electrical proximity; for instance, a coalition including bus 17 produces nearly the same impact as one with bus 18, meaning $\{17, 26, 39\}$ can be represented by the same signature as $\{18, 26, 39\}$. Applying this criterion alone reduces the 64 original coalitions to $3^3 = 27$ unique group-level signatures, since each cluster contains two ECs and can contribute 0, 1, or 2 ECs to a coalition. *Second*, ECs located in different feeders can be considered as independent. So, the ECs at buses 17, 18, 26, and 27 and ECs at buses 39 and 40 can be treated separately, assuming that there is no influence between these two sets of ECs. Combining this feeder separation with the clustering described above results in $3^2 = 9$ signatures for the first feeder and 3 for the second; removing the duplicated empty coalition yields a total of

TABLE V
COMPARISON OF SHAPLEY SAVINGS ACROSS DIFFERENT METHODS (\$) (IEEE 69-BUS).

Bus	Full Enum.	Sym.-only	Branch-only	Proposed Hybrid
17	80.606	80.713	80.597	80.737
18	80.633	80.713	80.624	80.737
26	102.910	103.100	102.894	103.994
27	103.480	103.100	103.469	103.994
39	0.012	0.012	-0.003	-0.003
40	0.012	0.012	-0.003	-0.003
Total	367.653	367.650	367.579	369.454

TABLE VI
COMPUTATIONAL EFFICIENCY ANALYSIS (IEEE 69-BUS SYSTEM).

Execution	Method	# Simulations	Time (s)
Sequential	Full Enumeration (Base)	64	719
	Symmetry-based only	27	304
	Branch-based only	19	215
	Proposed Hybrid (S+B)	11	126
Parallel	Full Enumeration	64	107
	Proposed Hybrid (S+B)	11	14.9

$9 + 3 - 1 = 11$ unique signatures. Thus, only 11 representative signatures are required instead of all 64 coalitions (Fig. 10).

We validate the proposed approximation by comparing the Shapley costs when all the coalitions are considered with the costs when only a reduced number of coalitions are considered. We compare three different versions which are (i) when only the symmetry-based criterion is used for the coalition reduction, (ii) when only the branch-based criterion is used, and (iii) hybrid when both (i) and (ii) are used. Table V provides a numerical comparison. As shown, the results obtained by the hybrid method are quite close to the case when full set of coalitions are simulated. Furthermore, Table VI summarizes the number of simulations and elapsed times for each method. As can be seen, the hybrid method is most computationally efficient.

d) Monte Carlo (MC) validation: To examine whether the observations from the preceding case studies generalize to other community configurations, we conduct a MC validation study. For each value of $|\mathcal{M}|$, which represents the number of ECs, $|\mathcal{M}|$ community locations are randomly selected from the network buses in each MC iteration, and the proposed signature-based Shapley reconstruction is performed using Algorithm 1. The exact Shapley values are computed by full enumeration and used as the benchmark.

In the screening step, we evaluate each single-EC case and all two-EC cases. Two ECs m and m' satisfy the single-EC criterion when the normalized difference between $v(\{m\})$ and $v(\{m'\})$ is no greater than 5%. Their marginal contributions are then compared when each is paired with every other EC k , and the normalized difference between $v(\{m, k\}) - v(\{k\})$ and $v(\{m', k\}) - v(\{k\})$ is no greater than 3%. ECs m and m' are grouped as similar when both criteria are satisfied. Branch-based reconstruction is applied when the interaction between different feeder branches is sufficiently small, using a relative tolerance of 3%. These criteria determine the representative coalitions used in the reconstruction.

Table VII reports the accuracy and computational performance of the proposed signature-based reconstruction for

TABLE VII
MONTE CARLO VALIDATION OF THE PROPOSED SIGNATURE-BASED RECONSTRUCTION (IEEE 69-BUS).

$ \mathcal{M} $	Samples	Error Count	Full Time (s/sample)	Signature Time (s/sample)	Mean Error (%)	95th Error (%)
5	100	500	50.3	25.6	0.675	1.614
5 [†]	100	500	50.3	28.3	0.915	1.488
6	100	600	99.8	48.8	0.658	1.693
7	100	700	199.2	88.5	0.696	2.037
8	30	240	437.2	161.9	0.880	2.093
9	30	270	907.2	300.5	1.027	3.126
10	30	300	1959.0	410.2	1.243	2.246

[†] EC capacities are randomly and independently varied within $\pm 10\%$.

different numbers of ECs. For each MC sample, the absolute allocation error is calculated for every EC and normalized by the grand-coalition value. The largest normalized error among the ECs is then defined as the sample-level error. The results show that the proposed method reduces the average computation time for all tested values of $|\mathcal{M}|$. The time reduction increases from approximately 49.1% for $|\mathcal{M}| = 5$ to approximately 79.1% for $|\mathcal{M}| = 10$. This trend occurs because, when $|\mathcal{M}|$ is small, there are fewer opportunities to form representative groups and reduce the number of coalition evaluations. Despite using the largest community-level error in each sample, both the mean and 95th-percentile errors remain low across all tested cases.

To further examine the effect of heterogeneous community sizes, we repeat the $|\mathcal{M}| = 5$ experiment using the same random seeds and the same EC locations, while randomly and independently varying the capacity of each EC within $\pm 10\%$. The corresponding results are reported in the second row of Table VII, marked with [†]. Under this setting, fewer ECs satisfy the screening criteria for grouping, resulting in a slightly smaller reduction in the number of coalition evaluations. Although the mean error increases from 0.675% to 0.915%, the 95th-percentile error remains comparable. These results show that the considered level of heterogeneity leads to more conservative grouping, with a slight decrease in coalition reduction and similar reconstruction accuracy.

3) IEEE 123-Bus and 9500-Node Test Systems

To further evaluate scalability and robustness, we apply the proposed scheme to larger networks such as the IEEE 123-bus and 9500-node distribution test systems.

a) Robustness of signature-based approximation on the IEEE 123-bus system: We consider six ECs located at buses 4, 6, 55, 59, 90, and 94 of IEEE-123 system. The six ECs include both identical and non-identical DER capacities to reflect diverse sizing scenarios. We assume that ECs at buses 4 and 6 are identical in size, forming a symmetric pair, whereas bus 55 has a 20% larger capacity than bus 59, while bus 94 has 50% more DER capacity than bus 90. This is intentionally done to test whether the signature-based method, which has a symmetry assumption, works when we have asymmetric DERs.

We address this case as follows. First, we treat the asymmetric pair (e.g., buses 55 and 59) as symmetric during coalition sampling and then distribute the final cost under Shapley allocation linearly with DER size. This uses the rationale

TABLE VIII
COMPARISON OF EXACT AND APPROXIMATED FINAL COSTS (SHAPLEY)
(IEEE 123-BUS SYSTEM).

Bus	Exact (\$)	Approx. (\$)	Abs. Error (\$)	Rel. Error (%)
4	12.12	11.72	0.40	3.30
6	12.12	11.72	0.40	3.30
55	41.34	41.99	0.65	1.57
59	50.17	50.39	0.22	0.44
90	24.97	24.46	0.51	2.04
94	36.24	36.69	0.45	1.24
Total	176.95	176.95	–	–

TABLE IX
SCALABILITY COMPARISON BETWEEN FULL ENUMERATION AND THE
PROPOSED HYBRID METHOD (IEEE 123-BUS SYSTEM).

Method	# Simulations	Time (s)
Full Enumeration	2^{30} ($\approx 1.07 \times 10^9$)	Intractable
Parallel Proposed Hybrid	359	1120

that communities with similar setups tend to have costs (or benefits) that are roughly proportional to their installed DER capacities. The results in Table VIII show that the approximation produces Shapley savings with reasonable accuracies.

b) Scalability on the IEEE 123-bus system: To further assess computational scalability, we consider a case with 30 communities leading to 2^{30} coalitions. The communities are assumed to be distributed across five weakly coupled branches of the IEEE 123-bus feeder, with 2, 4, 6, 8, and 10 communities per branch, respectively. Within each branch, communities with similar DER configurations are assumed to be paired in a two-by-two manner to form symmetric groups, enabling a signature-based approximation that aggregates coalitions with similar structural impacts. Table IX reports the resulting computation time for the IEEE 123-bus system. While exact Shapley value computation based on full enumeration is infeasible for 30 communities, the proposed hybrid signature-based method requires evaluating only 359 coalitions and completes within approximately 20 minutes on a standard laptop.

We emphasize that cost allocation is performed *ex-post*. This allows the proposed method to employ an approximate allocation for intermediate analysis, while higher-fidelity allocations can be conducted offline. Moreover, the total settlement amount across all ECs remains nearly identical to that of the full enumeration case, with only small error as demonstrated above. As a result, the proposed approximation is a practical and scalable solution without significantly degraded accuracy.

c) Monte Carlo (MC) validation on the 9500-node system: Finally, to examine the robustness of the proposed reconstruction under a substantially larger and different network topology, we conduct an additional MC study on the 9500-node distribution test system. For each of the 30 samples, five EC buses are randomly selected, and the reconstructed Shapley values are compared with the exact values obtained from all 32 coalitions. To limit the computational burden, this study is performed over four time periods ($|\mathcal{T}| = 4$). Across the 30 samples, the mean and 95th-percentile sample-level errors are 0.184% and 1.199%, respectively. The proposed method reduces the average computation time from approximately 822.0 to 250.1 s/sample. Compared with the five-EC cases on

the IEEE 69-bus system, the larger reduction can be attributed to the greater likelihood that randomly located ECs have weak or negligible interactions in the much larger network. Such cases are efficiently handled by the branch-based criterion, while symmetry-based grouping remains effective for ECs with similar marginal effects. These results confirm that the proposed reconstruction remains accurate and computationally effective for a substantially larger distribution network.

V. CONCLUSIONS

This work addressed the problem of fair cost allocation in energy communities, where the DERs across multiple communities jointly influence DLMPs and overall dispatch outcomes. The key challenge is to ensure fair distribution of operating cost savings among communities, accounting for their contributions to system-level performance. To this end, a bilevel optimization scheme was proposed from the perspectives of the CEA and the DSO. The bilevel problem was reformulated into a tractable single-level MILP using KKT conditions and strong duality. To ensure fair cost distribution among communities, the Shapley value was applied to reflect each community's marginal contribution to system-wide savings. The Shapley value method is applied as an ex-post settlement mechanism after the market clearing. The coordinated dispatch is first after solving the DLMP optimization scheme, then, the cooperative cost savings are allocated among the communities using the Shapley value.

Simulation results demonstrated that the proposed Shapley-based allocation method fairly reflects the locational impact and resource contribution of each community. Furthermore, sensitivity studies showed that communities hosting larger BESS units receive higher cost savings, highlighting the incentive compatibility of the proposed approach. We also proposed a coalition reduction approach, a signature-based method to enhance the computational scalability of the Shapley-value method on systems with large number of energy communities.

Future work will a scheme where the fairness scheme is directly incorporated into the DLMP optimization scheme. This would require an iterative framework linking Shapley-based allocations, community decisions, and DLMP outcomes. In addition, the proposed Shapley-based settlement is not intended to establish core stability. A systematic comparison with stability-oriented concepts will be explored.

APPENDIX A

DERIVATION OF THE LINEAR SINGLE-LEVEL OBJECTIVE
This appendix derives (10) from (2) using the strong duality of the lower-level DSO problem. Since the DSO problem is linear, its dual objective depends only on the constant right-hand sides of the primal constraints. In the present formulation, nonzero right-hand-side constants appear only in: (i) the bus injections $p_{n,t}^{\text{inj}}$, $q_{n,t}^{\text{inj}}$, and (ii) the upper and lower bounds on voltages and generator outputs.

Accordingly, the dual objective at time t can be written as

$$\begin{aligned} \theta_t(\cdot) = & - \sum_{n \in \mathcal{N}} \lambda_{n,t}^p p_{n,t}^{\text{inj}} - \sum_{n \in \mathcal{N}} \lambda_{n,t}^q q_{n,t}^{\text{inj}} + \sum_{n \in \mathcal{N}} \left(\lambda_{n,t}^{gp} p_n^g - \bar{\lambda}_{n,t}^{gp} \bar{p}_n^g \right) \\ & + \sum_{n \in \mathcal{N}} \left(\lambda_{n,t}^{gq} q_n^g - \bar{\lambda}_{n,t}^{gq} \bar{q}_n^g \right) + \sum_{n \in \mathcal{N}} \left(\lambda_{n,t}^u (v^{\min})^2 - \bar{\lambda}_{n,t}^u (v^{\max})^2 \right), \end{aligned} \quad (\text{A.1})$$

where $\lambda_{n,t}^p$ and $\lambda_{n,t}^q$ are the dual variables of the active/reactive power balances (5c)–(5f), and $\bar{\lambda}_{n,t}^{gp}$, $\bar{\lambda}_{n,t}^{gq}$, $\bar{\lambda}_{n,t}^u$ are the dual variables associated with the generation and voltage bounds. Using the injection definitions, $p_{n,t}^{\text{inj}} = -p_{n,t}^c$ for $n \in \mathcal{N}_c$ and $p_{n,t}^{\text{inj}} = -\hat{p}_{n,t}^{\text{load}}$ otherwise, and similarly $q_{n,t}^{\text{inj}} = -q_{n,t}^c$ for $n \in \mathcal{N}_c$ and $q_{n,t}^{\text{inj}} = -\hat{q}_{n,t}^{\text{load}}$ otherwise, the first two terms of (A.1) become

$$-\sum_{n \in \mathcal{N}} \lambda_{n,t}^p p_{n,t}^{\text{inj}} = \sum_{n \in \mathcal{N}_c} \lambda_{n,t}^p p_{n,t}^c + \sum_{n \in \mathcal{N} \setminus \mathcal{N}_c} \lambda_{n,t}^p \hat{p}_{n,t}^{\text{load}}, \quad (\text{A.2})$$

$$-\sum_{n \in \mathcal{N}} \lambda_{n,t}^q q_{n,t}^{\text{inj}} = \sum_{n \in \mathcal{N}_c} \lambda_{n,t}^q q_{n,t}^c + \sum_{n \in \mathcal{N} \setminus \mathcal{N}_c} \lambda_{n,t}^q \hat{q}_{n,t}^{\text{load}}. \quad (\text{A.3})$$

Equations (A.2)–(A.3) show that the DLMP payment terms in (2) appear explicitly within the dual objective. By strong duality, the primal DSO objective value at time t equals θ_t at optimality. Substituting the dual expression of θ_t into (2) and rearranging terms yields the linear single-level objective (10).

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